

# lxask

## Forgot the command? Just ask. Offline.

A 1.5B-parameter model that checks its own answers against the man pages, on your CPU, with no internet.

Aaron Thomas · Devin Hacks Kochi · Track 3: Works Anyway

github.com/arxhr007/lxask

# Looking up a command costs something

1. **Search or read man pages**
   You need the tool's name already, and often the internet.

2. **Ask a hosted AI assistant**
   Needs an account or API key, and your terminal context leaves the machine.

3. **Use a small local model**
   Private, but it invents flags that don't exist and never tells you.

*None of these work on a locked-down server, a plane, or an offline field laptop.*

# Plain English in, checked command out

```
$ lxask "which process uses port 3000, without lsof or netstat"
$ ss -ltnp 'sport = :3000'
```

**Explains** every part of the command

**Checks** each program and flag against the man pages and tldr

**Admits** what it can't confirm: "unverified"

**Never runs** anything; you read it first

# The demo runs with the constraint on

**What you will see**

**A restriction kept:**

"without lsof"

**A vague request:**

it asks back

**A doubtful flag:**

flagged "unverified"



Wi-Fi off

2 GB RAM limit

CPU only

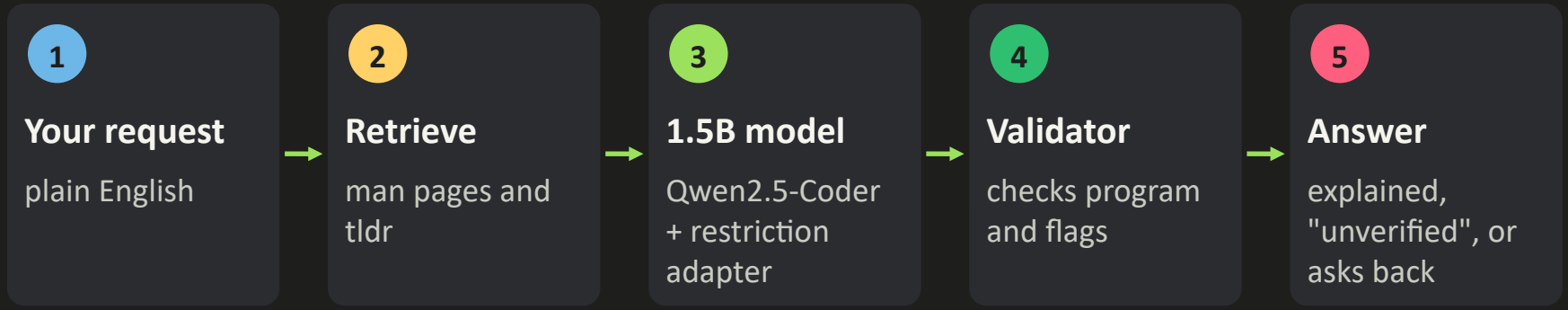
# Why it works anyway

| 1.1 GB | 1.3 GB | ~2 s | 0 |
| --- | --- | --- | --- |
| on disk | RAM at peak | per answer once loaded | accounts or keys needed |

**CPU only, no GPU.** No Ollama, no Docker, no root, no network after install.

**~7 s for the first answer, ~2 s after.** Linux on x86_64 and arm64 (arm64 tested under emulation).

# How it works

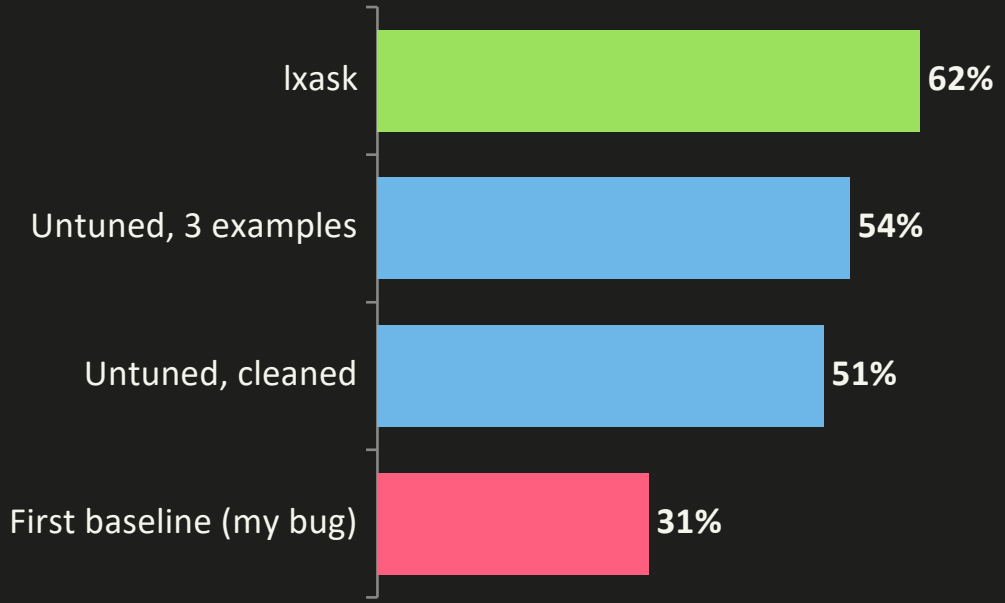



*If the validator objects, the model retries before answering.*

**Trained on** tldr-pages, NL2Bash and question-and-answer pairs written by language models (every command validated), then reinforcement learning on about 1,500 tasks whose commands are run in a sandbox.

# Measured on a public benchmark

NL2SH-ALFA: 300 shell tasks, each answer run in Docker. Share solved:




**+8 pts**

over a fairly prompted untuned model (paired test, p = 0.003)

**61% vs 54%** on the 237 tasks that don't overlap its training sources.

**The 31% was my own bug.** I found it, re-ran, and corrected it publicly.

GPT-4o scores 73% per the benchmark's authors; that figure was not re-run here.

# Limits, next steps, and an ask

## Honest limits

- Linux only, English only
- Misses by a detail: a flag, a field number
- Not a replacement for knowing your system

## Next

- macOS support
- A self-check command, lxask doctor
- Packages for Arch, Homebrew, Nix

**Try it, and tell me where it's wrong**

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